Uses of agar:

- It is one of the principle ingredients in the preparation of solid or semisolid media.
- It is used as a solidifying agent in culture medium.
- It is extracted from certain seaweeds belonging to genera of red algae like *Gelidium* and *Gracilaria* (Figure 5.1).

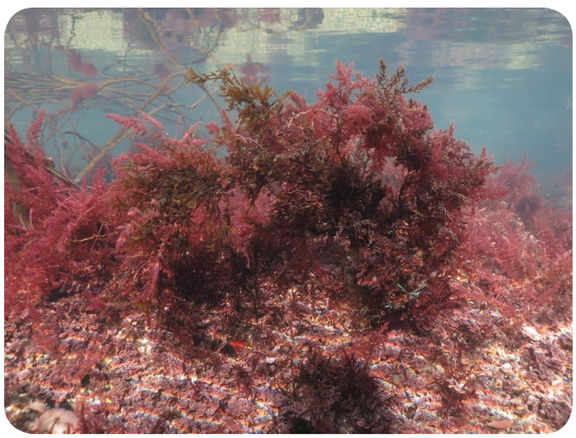


Figure 5.1: *Gelidium* – Red algae

- It is a sulphated polymer mainly consisting of D-galactose.
- Agar is a highly preferred solidifying agent because it does not affect the growth of microorganisms. Agar is also used in the food and pharmaceutical industries.
- The purified form of the agar is called Agarose. It is prepared by removing the pectin from the Agar. It is used in Molecular Biology laboratory for the separation of DNA molecules by gel electrophoresis.



Agar was first described for use in Microbiology in 1882 by the German microbiologist Walther Hesse, an assistant working in Rober Koch's laboratory, as suggested by his wife Fannie Hesse.

A cheap substitute for agar in microbial culture media is Guar gum, which can be used for the isolation and maintenance of thermophiles.

HOTS

Why is agar preferred to gelatin as a solidifying agent in culture media?

5.2.1 Physical Nature of Agar Medium

The concentration of agar plays a major role in determining the consistency of the medium. A medium with agar concentration of 2% or greater is said to be a solid medium and that of 0.5% is said to be a semisolid medium (jelly like appearance). Tabel 5.2 lists the concentration of agar in

Table 5.2: Concentration of agar in media

Nature of Medium	Concentration	Example	Uses
Solid	2%	Nutrient agar	To isolate microorganisms on petridish and forming agar slant
Semisolid	0.5%	SIM (Sulphur Indole Motility medium)	Agar stab to observe motility
Liquid	0%	Nutrient broth	To observe biochemical reaction.

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